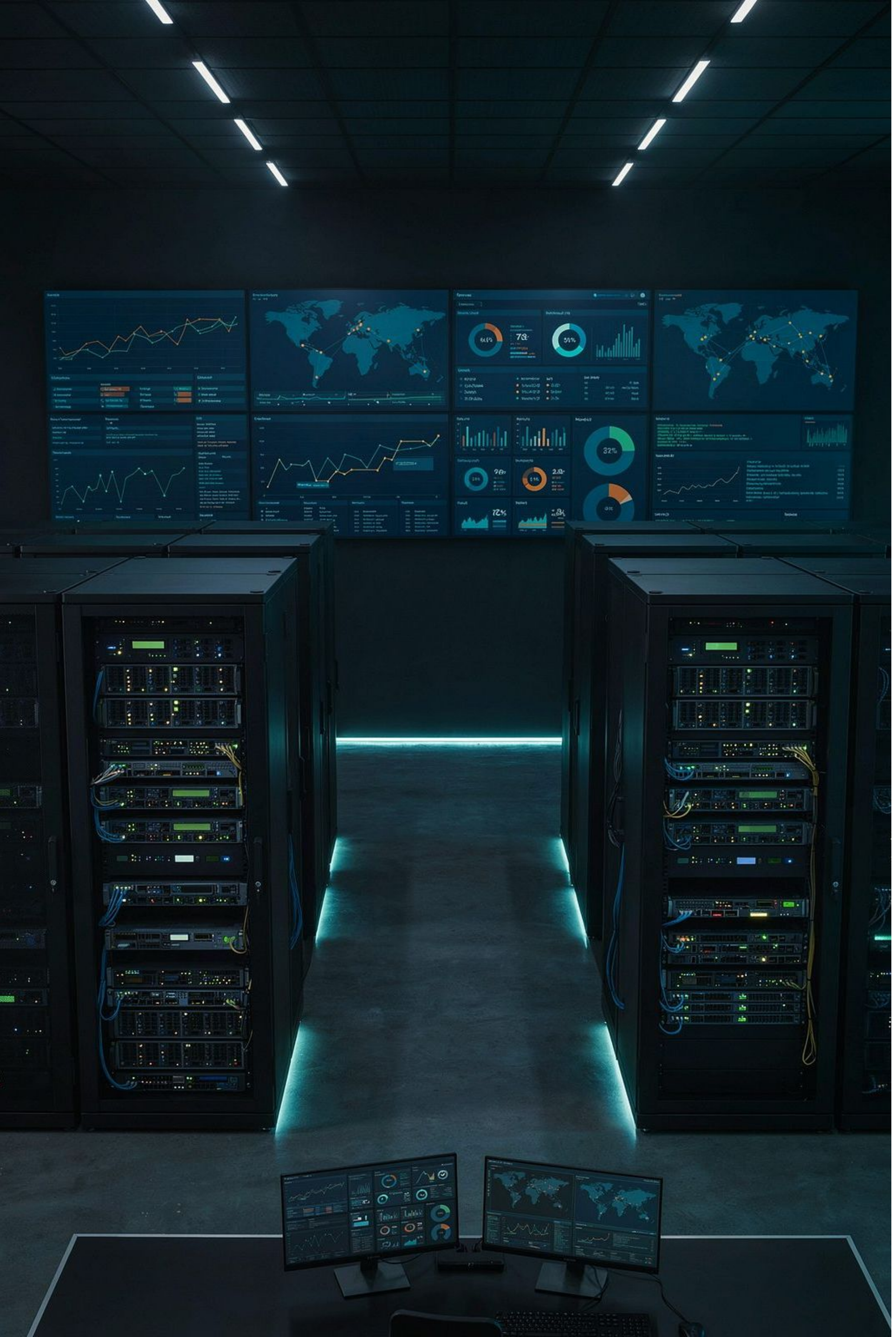


12-Week Enterprise Engineering Program

From Fundamentals → Enterprise-Ready Engineering

12

Weeks

36

Sessions

3

Hours / Session

108

Total Hours

Central Philosophy: We are not learning technologies independently — we are learning how they work together in a real enterprise environment.

Our 12-Week Learning Journey

1

Phase 1–2

Engineering Foundations: OS, Networking, Agile, SQL, DevSecOps

2

Phase 3–5

App Engineering: Python, Java, Spring Boot, Microservices, AWS

3

Phase 6

AI Engineering: LLMs, Prompt Engineering, RAG, AI-assisted Security

4

Phase 7

DevOps & Quality: GitLab, Terraform, Splunk, Playwright, Capstone

What Is Enterprise Software Engineering?

Enterprise software supports large organizations — their processes, customers, transactions, data, and regulatory requirements. Examples: banking platforms, mortgage systems, ERP, healthcare, e-commerce, and government systems.

Scalable

Handles growing users and transactions

Secure

Protects sensitive business data

Reliable

Operates even when components fail

Observable

Provides logs, metrics, monitoring

Compliant

Meets regulatory requirements

Product vs. Project Delivery

Project Thinking

Delivers a defined scope with a clear endpoint.

Example: Build a mortgage delinquency reporting system.

- Defined scope, timeline, budget
- Completion milestone
- Start → Build → Deliver → Complete

Product Thinking

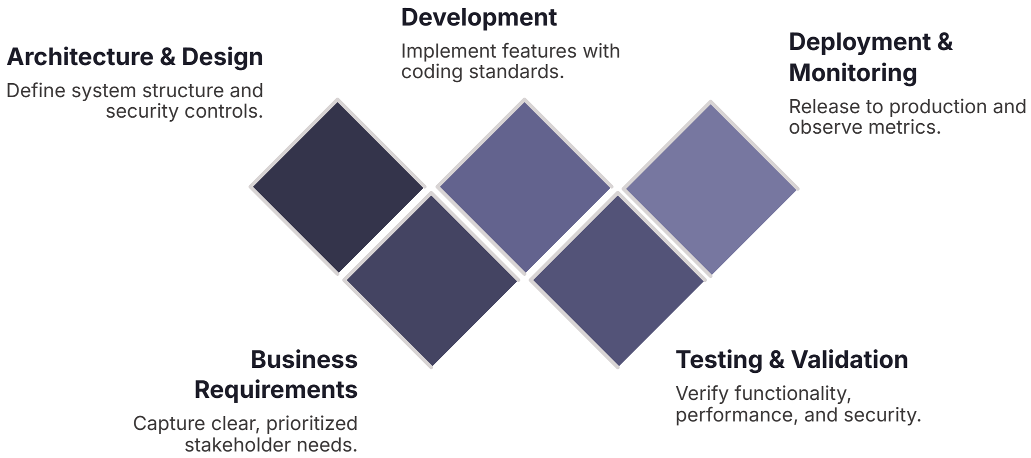
Continuously evolves through versions.

Example: Mortgage Intelligence Platform

- V1 → Mortgage Management
- V2 → Delinquency Monitoring
- V3–5 → Risk Analytics, AI, Fraud Intelligence

Discover → Build → Release → Measure → Improve → Repeat

SDLC & Path to Production



Path to Production

Developers don't write code and deploy directly to production. A typical enterprise flow includes:

Without a structured lifecycle, requirements are misunderstood, security is missed, and production failures increase.

01

Jira Story → Development → Git Commit

02

Pull Request → Code Review → Automated Testing

03

Security Scanning → QA → UAT

04

Release Approval → Production

Enterprise Environments & Governance

LOCAL

Developer workstation — development, debugging, unit testing

DEV

Shared environment — integration and developer testing

TEST / QA

Used by testing teams for quality validation

UAT

Business users verify requirements are satisfied

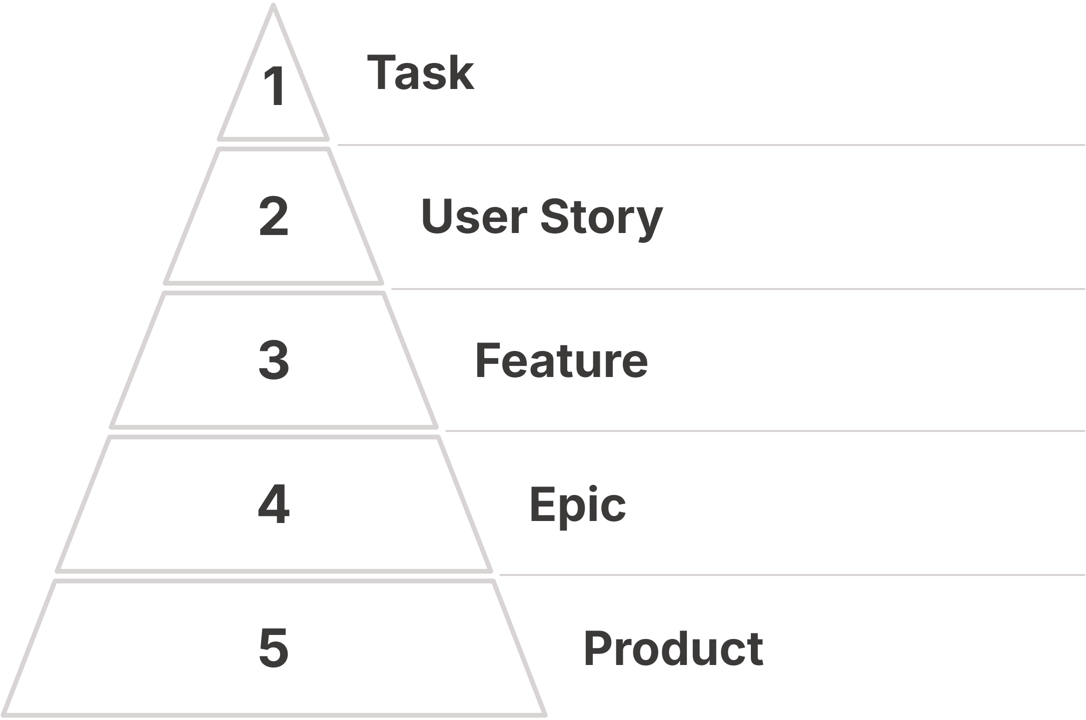
PROD

Live system used by real customers and business teams

Governance reduces production failures, security incidents, unauthorized changes, and compliance violations. It includes architecture reviews, security reviews, change management, and audit evidence.

Agile Delivery & Team Structure

Agile Hierarchy



Cross-Functional Team

- **Product Owner**

  Defines priorities and business value

- **Architect**

  Defines architecture and major technical decisions

- **Developer / QA / Security**

  Build, validate, and secure the application

- **DevOps / SRE**

  Support CI/CD, infrastructure, and reliability

The Mortgage Domain

Technology solves business problems. Understanding the domain is as important as knowing the technology. Our program uses **Fannie Mae–style secondary mortgage scenarios** as the enterprise context throughout training.

Origination

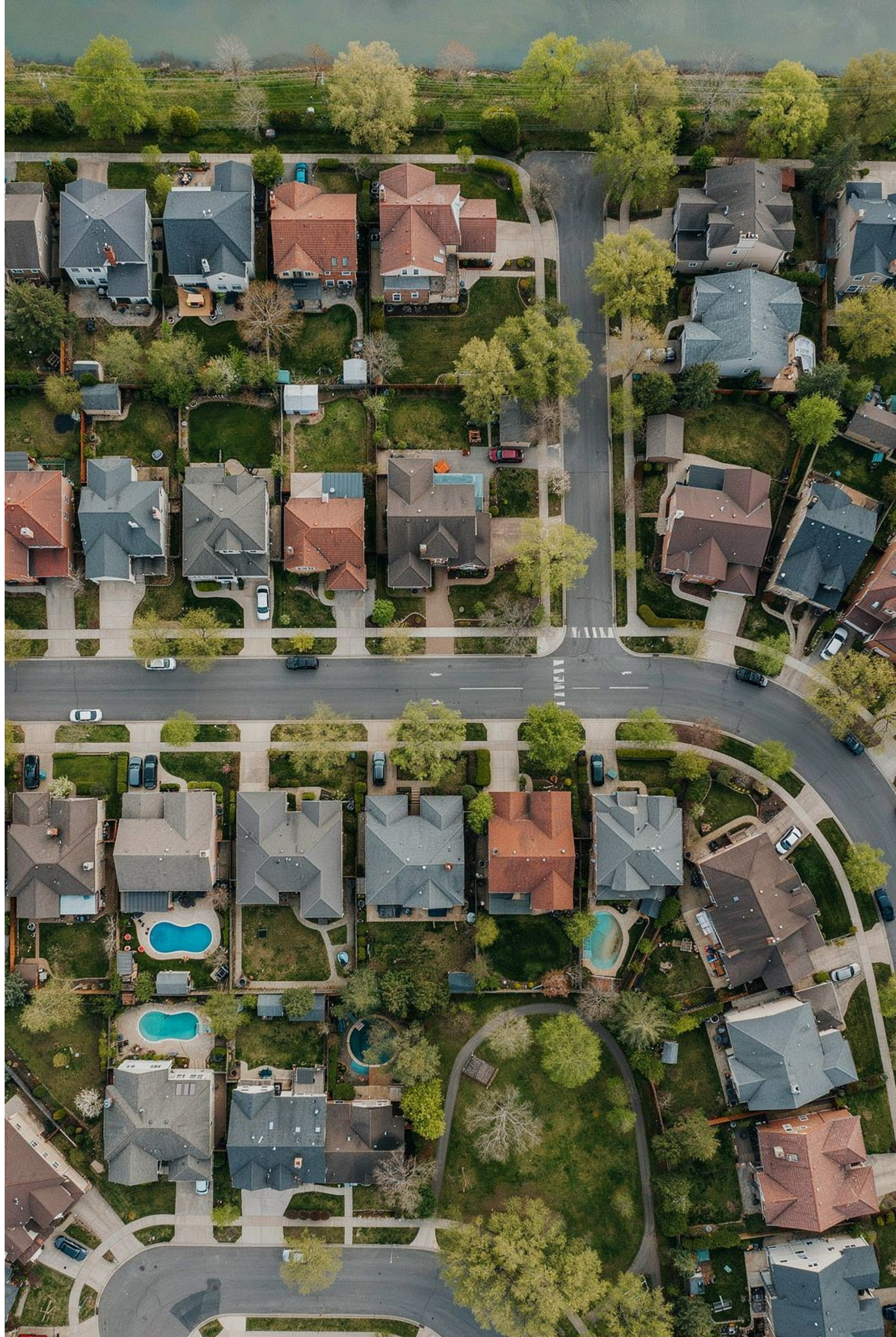
Borrower applies → Verification → Underwriting → Approval → Closing → Active Loan

Primary & Secondary Market

Lender → Mortgage Acquisition → Securitization → Servicing Intelligence

Servicing & Delinquency

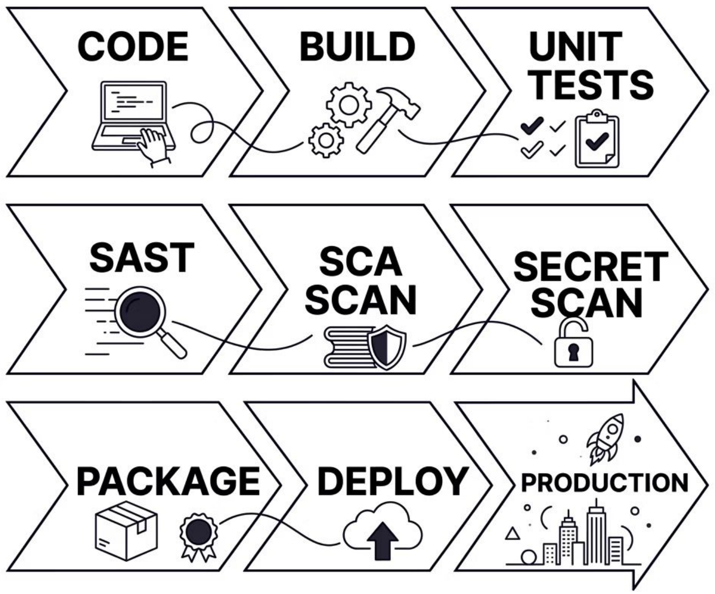
Payment processing, balance tracking, delinquency monitoring, alerts, and analytics



Cybersecurity & DevSecOps

Why Security Matters

Mortgage platforms process sensitive borrower and financial data. Security cannot be added only after development.

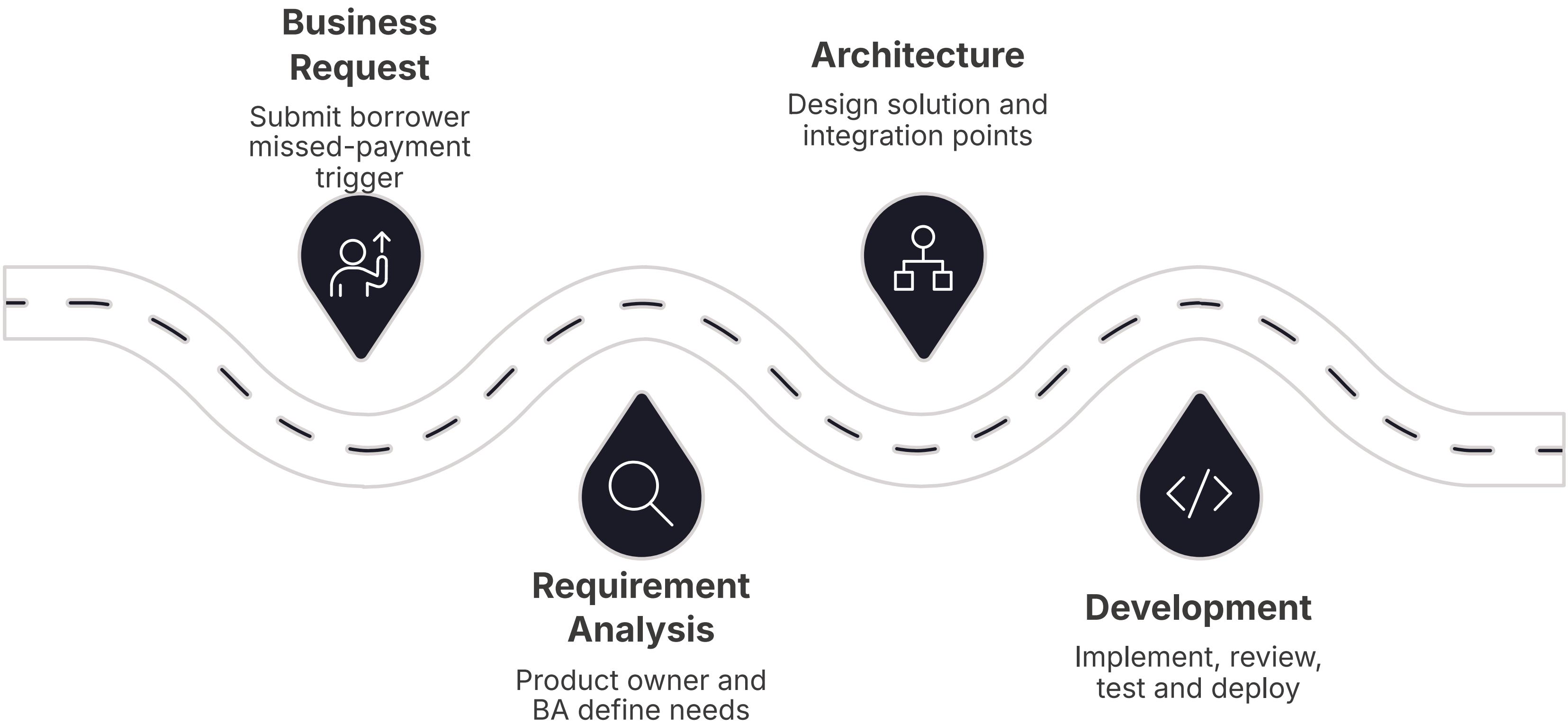


Confidentiality Only authorized users access data	**Integrity** Data is not changed improperly	**Availability** Systems remain accessible to legitimate users

Key cybersecurity roles include: Application Security Engineer, Cloud Security Engineer, SOC Analyst, Penetration Tester, IAM Engineer, DevSecOps Engineer, Security Architect, and GRC.

Hands-On: Business Request → SDLC

> **Business Requirement:** When a borrower misses a mortgage payment, the platform should identify the situation and notify the servicing team.



Stakeholders involved: Product Owner, Business Analyst, Servicing Team, Architect, Developers, QA, Security, DevOps, and Operations/SRE.

Key Learning: Enterprise engineering involves far more than writing code — it spans governance, security, testing, and continuous monitoring.